



Practical Stereophotoclinometry for Modeling Shape and Topography on Planetary Missions

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Received 2021 September 10; revised 2021 December 13; accepted 2021 December 23; published 2022 May 10

Abstract

Stereophotoclinometry (SPC) is a technique to extract topographic information from images acquired by spacecraft. It combines stereophotogrammetry and photoclinometry to produce a product that has the accuracy of stereo with the resolution of photoclinometry without the restrictions common to both. We describe the implementation of this technique in the context of digital terrain model (DTM) generation for a small-body mission. We detail the process and the data used to generate SPC-derived DTMs at progressively increasing resolutions. The highest-quality DTMs are generated using four images optimized for topography, a 30° emission angle with the emission azimuth (spacecraft position) to the north, east, south, and west of the target, and one image optimized for albedo (a low incidence angle such that most of the image pixels' digital numbers are based upon albedo rather than topography). We discuss implications for mission planning and how SPC-based DTM generation can support spacecraft navigation. As a case study, we share outcomes from the modeling performed for the OSIRIS-REx mission to asteroid Bennu.

Unified Astronomy Thesaurus concepts: [Asteroids \(72\)](#)

1. Introduction

Stereophotoclinometry (SPC), a fusion of stereophotogrammetry and photoclinometry, is used to create image-based digital terrain models (DTMs; Gaskell 2001, 2004; Gaskell et al. 2008). It leverages the strengths of each technique and mitigates the weaknesses of the other. Stereo provides absolute 3D position, while photoclinometry generates a higher spatial resolution (Palmer et al. 2016). Photoclinometry provides high-resolution topography through pixel-to-pixel slope estimation, while stereo anchors this topography in 3D space. Additionally, SPC solves a system of three equations for every pixel, resulting in the determination of not only the two orthogonal surface slopes but also albedo (R. W. Gaskell et al. 2022, in preparation). Because the resulting model contains both albedo and topography, SPC can use that information to render a synthetic image that aids in direct comparisons to original input images.

The suite of tools known as SPC was developed by Robert Gaskell (Gaskell et al. 2008) at the Planetary Science Institute (PSI) in support of optical navigation efforts. The SPC software hosted by PSI, which we use here, has evolved over the years to support not only optical navigation but also high-resolution DTM generation. It comprises 87 modules and 87,000 lines of FORTRAN code. The system runs using the command line of Unix operating systems. Information on software licensing and an extended user's manual can be found at <https://spc.psi.edu>. In this work, the names of the most common modules are included in the text with brackets around them to avoid confusion with the prose itself.

To determine the topography of the targets of planetary missions, SPC has been used extensively (e.g., Table 1; see also Barnouin et al. 2020). Recently, it was used for scientific terrain modeling (Barnouin et al. 2019) and optical navigation (C. D. Adam et al. 2022, in preparation) by the Origins, Spectral Interpretation, Resource Identification, and Security–Regolith Explorer (OSIRIS-REx) sample return mission (Lauretta et al. 2017, 2021) to asteroid (101955) Bennu. The OSIRIS-REx mission provided an exceptional opportunity for understanding the performance of SPC; extensive ground-based testing was performed (Barnouin et al. 2020; Craft et al. 2020, E. E. Palmer et al. 2022, in preparation, Weirich et al. 2022), followed by flight validation using laser altimetry (Daly et al. 2017; Seabrook et al. 2019; Al Asad et al. 2021, J. A. Seabrook et al. 2022, in preparation) and other image-based terrain modeling techniques (Al Asad et al. 2021). The demonstrated performance has increased confidence in SPC and improved understanding of the best practices for observational planning, in-flight validation, strengths and weaknesses, and expected performance.

Here we apply this improved understanding to detail the approach for using the SPC software suite during a space mission. The emphasis is on application to a rendezvous mission, as procedures were developed and lessons were learned in this context, but, as shown in Table 1, SPC has utility on planet-sized bodies as well, and the same principles apply to those missions. We first introduce the components and nomenclature of SPC, then describe the process and operational practices needed to make SPC work, and finally discuss its application with OSIRIS-REx as a case study.

It is important to note that the focus on these procedures is effective for a rendezvous/orbital-focused mission. Additionally, we provide guidelines for an optimized set of observing



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Table 1
Objects that Have Been Modeled Using SPC

Object	Type	Shape	Mission	Scope	Reference
Eros	Asteroid	Triaxial	NEAR	Global	Gaskell 2008
Itokawa	Asteroid	Triaxial	Hayabusa	Global	Gaskell et al. 2008
Vesta	Asteroid	Spheroidal	Dawn	Global	Gaskell et al. 2011
Ceres	Asteroid	Spheroidal	Dawn	Global	Park et al. 2019
Ryugu	Asteroid	Spheroidal	Hayabusa2	Global	Watanabe et al. 2019
Mathilda	Asteroid	Spheroidal	NEAR	Regional	Weirich et al. 2019
Bennu	Asteroid	Spheroidal	OSIRIS-REx	Global	Barnouin et al. 2019; Palmer et al. 2019
Lutetia	Asteroid	Spheroidal	Rosetta	Regional	Jorda et al. 2013
Steins	Asteroid	Spheroidal	Rosetta	Regional	Jorda et al. 2012
67P/C-G	Comet	Complex	Rosetta	Global	Gaskell et al. 2014
9P/Tempel 1	Comet	Spheroidal	Deep Impact, Stardust	Global	Ernst et al. 2015
Dione	Icy moon	Spheroidal	Cassini	Global	Gaskell 2013a
Tethys	Icy moon	Spheroidal	Cassini	Global	Gaskell 2013d
Mimas	Icy moon	Spheroidal	Cassini	Global	Gaskell 2013b
Atlas	Icy moon	Complex	Cassini	Global	Daly et al. 2018; Martin et al. 2020
Calypso	Icy moon	Complex	Cassini	Global	Daly et al. 2018; Martin et al. 2020
Epimetheus	Icy moon	Complex	Cassini	Global	Daly et al. 2018; Martin et al. 2020
Helene	Icy moon	Complex	Cassini	Global	Daly et al. 2018; Martin et al. 2020
Janus	Icy moon	Complex	Cassini	Global	Daly et al. 2018; Martin et al. 2020
Pan	Icy moon	Complex	Cassini	Global	Daly et al. 2018; Martin et al. 2020
Pandora	Icy moon	Complex	Cassini	Global	Daly et al. 2018; Martin et al. 2020
Prometheus	Icy moon	Complex	Cassini	Global	Daly et al. 2018; Martin et al. 2020
Telesto	Icy moon	Complex	Cassini	Global	Daly et al. 2018; Martin et al. 2020
Phoebe	Moon	Spheroidal	MRO	Global	Gaskell 2013c; Ernst et al. 2018
Deimos	Moon	Spheroidal	Viking, MGS, MRO	Global	Ernst et al. 2018
Phobos	Moon	Spheroidal	Viking, MGS, MRO	Global	Gaskell 2011; Ernst et al. 2018
The Moon	Moon	Spheroidal	Various	Regional	Palmer et al. 2016; Weirich et al. 2019
Earth	Planet	Spheroidal	Various	Regional	Palmer et al. 2016
Mercury	Planet	Spheroidal	MESSENGER	Global	Gaskell et al. 2011

conditions for taking images, although, as seen in Table 1, SPC can make quality DTMs in a wide range of applications.

2. SPC Nomenclature

A variety of data constructs and representations are used by SPC. These terms are defined and discussed below.

2.1. Ground Sample Distance

Ground sample distance (GSD) describes the spatial separation of each element in a gridded DTM product. It is not necessarily equivalent to the resolution of the information input into a DTM but is a measure of the spatial separation between each point in the DTM. For example, a high-resolution model with a small GSD can be oversampled such that its GSD is smaller than the resolution of the data input into the model. Thus, in general, the GSD provides information on the length scale of each data element but not the actual scale (resolution) of the information content in the DTM.

2.2. Maplets

The most basic SPC-specific data structure is a maplet, which is a small, square, 2D grid at a defined GSD containing shape and albedo information. Usually 99×99 pixels, the maplet is located in 3D space by a vector from the center of mass of the object to the center of the maplet. A plane normal to the surface defines the orientation of a 2D grid with consistent element spacing. The data at each grid element are the topographic height, defined as the height deviation from the

normal plane, and an albedo value (see R. W. Gaskell et al. 2022, in preparation, for mathematical definition).

As part of the SPC process, “subimages” are constructed for each maplet. Subimages are a resampling of a source image over the size of the maplet (discussed further in Section 3.4.2).

A regional or global DTM consists of maplets that cover the surface with sufficient overlap to cover every pixel. Hundreds to thousands of maplets can be used depending on the chosen GSD and spatial extent. Once a suite of overlapping maplets is determined, they can be resampled into a gridded product, such as a plate model, vector/facet model, point cloud, or 2D gridded DTM.

2.3. Landmarks

For every maplet, there is a landmark with its location defined by a stereo solution. By definition, the landmark itself contains the aforementioned vector that defines the center point of the maplet. It can be thought of as a control point, but, as implemented in the SPC software suite, the landmark file also contains additional metadata about the maplet that it references. These metadata include the GSD, the half-width of the maplet (in pixels), the normal plane and orientation of the 2D grid, and a calculated value of the accuracy of the position. The image information stored is a reference to every associated image and the pixel/line position in each image of the maplet. It also includes each image that has this maplet on its limb. Finally, it includes any overlapping maplets and their relative position.

2.4. Bigmaps

A bigmap is a larger version of a maplet that is derived from resampling the data from many maplets in the region (R. W. Gaskell et al. 2022, in preparation). Higher-resolution maplet data are given more weight when creating the bigmap. A bigmap can be up to 5000 pixel elements wide and, like a maplet, always contains an equal number of grid elements in the two orthogonal directions.

A bigmap is one of the major products that is distributed for scientific or operational use, as individual maplets describe too small an area for most purposes. The bigmap provides a regional DTM in a 2D grid that includes a height defined as a displacement from the normal plane and a scaled albedo value. These regional DTMs can be converted into formats usable by standard geographic information system tools or used in computational models.

2.5. Center of Mass versus Center of Figure

The origin of the body-fixed Cartesian coordinate system is defined by SPC to be at the center of mass. When the spacecraft position, and thus camera position, is translated to a body-fixed frame, the positions are relative to the center of mass.

During approach, there is no ability to differentiate between the center of mass and center of figure, and as such, navigation is based solely upon the center of figure. However, when the spacecraft is in orbit, radio tracking data can define the offset between the center of figure and center of mass. Thus, when the spacecraft has not reached the object, it is assumed that the center of figure is the same as the center of mass. Images will be registered to the center of figure until an offset is determined, either by SPC or the navigation team, in which the navigation reconstructed position switches to center of mass.

2.6. Albedo

For every pixel in a maplet, SPC generates an albedo value. The albedo value reported is the geometric albedo at a phase angle of 0° with the opposition surge ignored (Hapke 1981). These albedo values are normalized to a relative scale between zero and 200 and stored as an integer in a single computer byte to reduce file space (a legacy implementation). Some work has been done using this albedo value as a proxy for Bond albedo (Palmer et al. 2013), but the main benefit of including the albedo value is to provide renderings of the SPC-derived models that more realistically match the images while developing the SPC products, and later for their use in scientific analyses and spacecraft navigation.

2.7. Residuals

A tool in the SPC software suite [RESIDUALS] generates a metric that is used to evaluate the self-consistency and quality of the model. This metric is a measure of how closely each maplet or image aligns with the complete model, which is computed as the sum of the squares of the deviations. It provides a quantitative measure of the quality of a model through the consistency of the model with underlying data. However, it does not indicate if input data are missing or of poor quality. Its main use is to assess alignment errors between the image and the model to indicate how consistent the model is with the data; it also indicates where manual interventions

might be required and when no further processing is required because the model is complete. This quantitative measure is correlated with rms topographic error (E. E. Palmer et al. 2022, in preparation; Weirich et al. 2022).

2.8. Digital Numbers

Images collected by spacecraft are digital products consisting of matrices of pixels that correspond to locations on a focal plane. Each pixel has a value or digital number (DN), which is the energy measured by the pixel and interpreted as the brightness of the image on the location referenced by that specific focal plane element. The downlinked image, therefore, is a 2D array of values that translates into a gray-scale image.

The range of DN values is chosen to match the dynamic range that the focal plane array can accommodate. They typically range from 8 to 16 bits. As an example, the cameras used for SPC on the OSIRIS-REx mission have DN ranges of 12 (e.g., the navigation camera NavCam 1; Bos et al. 2018, 2020) or 14 bits (the OSIRIS-REx Camera Suite, OCAMS; Rizk et al. 2018; Golish et al. 2020).

Images that are as-downlinked are known as “raw” or level 0 data. These images have no corrections for instrument bias, artifacts, or read noise. They frequently undergo various calibration steps (Bos et al. 2020; Golish et al. 2020), either before they are ingested into SPC or as a part of the ingestion process.

There are several important factors that can impact the quality of the images used in the terrain modeling process. If the exposure time is too long, the DN values can either be clipped (truncated at the maximum value) or be in a region where the detector provides a nonlinear response for the input. Short exposures can result in the range of DN values being very small, thereby losing variations in brightness. Wide point-spread functions result in pixels not being a single point but a composition of numerous neighboring pixels. Camera distortion is an issue that is typically handled by distortion parameters within the mission-specific software for image ingestion to remove those effects. High-quality calibration can fix some of these issues.

While there are many potential camera artifacts that can impact the quality of the images, the ability of SPC to use a large number of images through a wide range of observing geometries allows these errors to be reduced. Multiple observations allow the system to be overconstrained, in which the key variables can be solved for by finding the solution that has the best fit to the data.

3. Method

3.1. An Overview of SPC

The SPC process is initialized with a crude starting DTM, which may be informed by premission telescopic data or simply a sphere, and related planetary constants, such as pole position and rotation rate, to support early model convergence. The process can be broken into two sequential parts: limb-based modeling and maplet-based modeling (Figure 1). The first part, limb-based modeling, is used when approaching the target where individual images capture a large fraction of the target. Model points on the limb derived from the images are used to refine the fidelity of the DTM. As the body begins to fill the field of view, many images do not contain limb views, and a switch to maplet-based modeling is necessary.

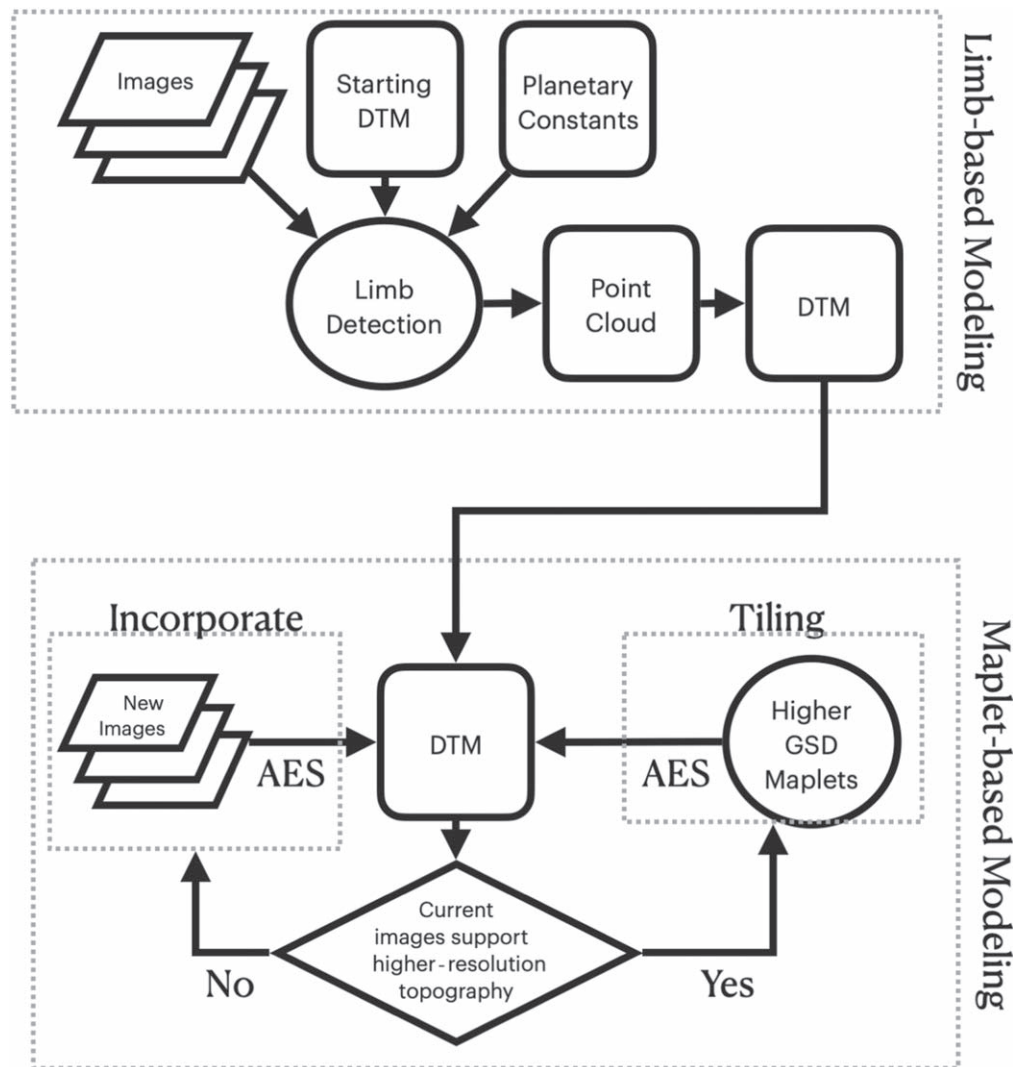


Figure 1. High-level view of the stages of building a DTM using SPC. A starting DTM and an estimate for the object’s rotation rate and pole (i.e., planetary constants) are required. The starting DTM is improved using limb vectors during the spacecraft’s approach to the target. Once the target fills the field of view, maplet-based modeling is better. First, SPC covers (or tiles) the surface with small 2D maplets and extracts topography from the position and shading of the images. Second, new images are added to the system, allowing higher-resolution data to improve the DTM. These two processes are repeated until the model reaches the desired GSD.

In the maplet-based modeling process, images are ingested and aligned (or registered) to the refined DTM. Then, maplets are created so that they cover and overlap the surface. Each image that intersects a maplet is assigned to that maplet and aligned to the DTM. The maplet’s exact position in 3D space is calculated by a weighted triangulation of the images (a full discussion is in Section 4.2). The topography of each vertex element of a maplet is calculated predominantly by photoclinometry. As higher-resolution images are integrated into the process, higher-resolution maplets are created. This process is continued until the desired resolution is reached and the iterative SPC processing has minimized errors through minimization of the formal uncertainty (Figure 1). More details of these steps are provided in subsequent sections.

3.2. Initialization Parameters

The quality and fidelity of the starting DTM can be extremely low, but the closer it is to the actual surface shape, the faster the model will converge. The starting DTM is used for two purposes. The distance of the surface from the center of

mass provides the geometry used to project and coregister the images. The starting DTM is also used to load each maplet with initial topography. Preloading lower-resolution DTMs into the maplets allows for orthorectification of the source images and decreases the number of iterations required for convergence.

There are many possible sources of low-resolution DTMs that can be used to start an SPC model. One of the most common is light-curve inversion (Ostro & Connelly 1984), as has been done for the comet 67P/Churyumov–Gerasimenko (67P C-G; e.g., Kelley et al. 2009) and other small bodies. Using inverse techniques and assuming a constant albedo, one can identify a global shape such that the variations in observed brightness match what the physical shape would be. Inversion cannot model concave shapes, but it provides a reasonable starting point when no other data are available. Analysis of the light curves can also help estimate the size of the object, its rotation rate, and its pole orientation (planetary constants).

Alternatively, the starting shape can come from radar images (e.g., Nolan et al. 2013). Bodies that come close to Earth can be imaged by radar instruments, and, from these data, one can estimate the shape, size, and pole of an asteroid. This technique

is typically more accurate than light-curve inversion, but it is limited to asteroids that pass close enough to Earth to obtain sufficient radar returns.

Of course, if a planetary mission has visited an object before, then it is likely that there will be a preliminary DTM of the object. These models can also provide an excellent starting point for SPC. They can come from stereophotogrammetry, laser altimetry (lidar), or limb-based techniques (e.g., Stooke & Pajola 2019). Along with the basic shape of the asteroid, the size and coordinate system are important to properly reference new images to the model and each other consistently. The rate of rotation, pole position, prime meridian, and size of the target constitute key components.

The orientation of the pole is a critical element of a DTM. If the pole location R.A. and decl. are incorrect, inconsistencies in the model will arise from misaligning the images with the model based on the spacecraft position, pointing, and time. Errors in the pole position can result in poor topography and the appearance of artifacts, such as an unphysical pole wobble. This pseudowobble will result in a larger volume for the model and cause the features on the surface to be misaligned when compared to images taken at different times.

As the spacecraft approaches an object, it is critical to have an accurate rotation rate. If the rotation rate is incorrect, then the image will be misaligned with the model. This misalignment will result in a disagreement between different epochs of data. Not only will the prediction of the position of features be offset from physical reality, but this misalignment will increase with time. If the image data are used to update the spacecraft's position, it will result in an in-track error.

Although the prime meridian is not an intrinsic property of the asteroid, it is, along with the pole, a key component of a geographic coordinate system. If there has not been a prime meridian published in the literature, then a feature must be selected by the mission team to use as the prime meridian (Archinal et al. 2011). Regardless of how the prime meridian is defined, all data must have the same definition to ensure consistency and the highest-fidelity topography.

The pole, rotation rate, and prime meridian are maintained in a planetary constants kernel (PCK) that is loaded into SPC [MAKE_SUMFILES] from the SPICE framework (Acton 1996; Acton et al. 2018). The navigation team that generates the position and pointing kernels must also have the same PCK or the data will not be self-consistent. Additionally, when there is an improvement in the knowledge of the PCK values, care must be taken to adjust both the navigation kernels and the internal SPC data accordingly. Finally, as the target's state information is improved, these updated parameters are fed back into the system to reduce error. There are basic SPC tools for identifying an updated solution for the pole position and rotation rate where it calculates an R.A./decl. and rotation rate to minimize the error in position between the maplets and the images.

The last parameter that should be determined is the size, which will be improved as better data become available and are integrated into the DTMs. If the general size of the target is incorrect, then the correction to the navigational position for the spacecraft will also be incorrect. For example, if the DTM's size is too large to fit the image data, the spacecraft will appear to be closer to the target than it actually is. This problem is known as the size degeneracy, where, with typical data, one cannot determine the size versus distance by image data alone.

This degeneracy can be broken by using spacecraft telemetry and Doppler tracking data through the telecommunication system to determine the actual size of the object. During approach at low phase, the line-of-sight speed vector of the spacecraft is one of the best known navigational parameters (through Doppler). Although the position of the target may be uncertain, taking a set of images every few days will provide the data to break the size/distance degeneracy. And although the distance between the target and the spacecraft has large uncertainties, the change in position for both can be calculated to high precision. Taking both the well-known time and distance traveled, it is possible to converge on a size solution.

3.3. Approach

3.3.1. Distant Approach: Center Finding and Limb-based Topography

Here we start with a basic DTM as described in Section 3.2 as we approach the object. This initial model is not much more than a placeholder but is still needed for navigation and rotation parameter determination (pole orientation and rotation rate).

During distant approach, when the shape of the target is unresolved, the navigation team identifies the center of brightness of the target to subpixel accuracy. However, as the object becomes resolved, the difference between the center of brightness and center of figure becomes more significant, depending on the approaching phase angle. The navigation team applies a center of brightness-to-center of figure correction to the measurement by computing a predicted offset based on the best available DTM or triaxial ellipsoid (Jackman et al. 2016; Antreasian et al. 2019). Errors in the a priori DTM can introduce bias in the approach navigation targeting. For example, during the Rosetta mission to 67P C-G, the comet's actual shape was very different from the shape that was identified from light curves (Kelley et al. 2009). As such, a new shape had to be generated that included the significant asymmetry of the object (Jorda et al. 2016).

3.3.2. Limb-based Modeling

When an object becomes spatially resolved, a shape can be generated using observation of its limbs during rotation (Figure 2). Numerous images are taken during a single rotation with the cadence chosen to provide the required model resolution. For example, a cadence that corresponds to every 20° rotation provides a limb point every 20° on the surface, resulting in 18 images; in the case of Bennu, that translates to ~87 m per point along the equator. Each limb point can be projected into 3D space [LIMBER] with the assumption that the limb point is orthogonal to the viewing geometry and located in plane to the center of mass. Each limb point becomes part of a point cloud that can be meshed to make a global DTM. For a flyby mission when the rotation is too slow to get limb profiles on the entire object, a limited DTM can be generated that constrains only a portion of the object.

From testing and operational experience, we have found that updating the starting model with a limb-based model provides an improvement over typical starting models when the object is 25–50 image pixels in diameter or better. Higher-resolution limb-based models can be generated as the resolution of the images improves and rotation imaging is repeated. As previously discussed, the highest-quality starting model for SPC reduces processing time, error, and size bias.

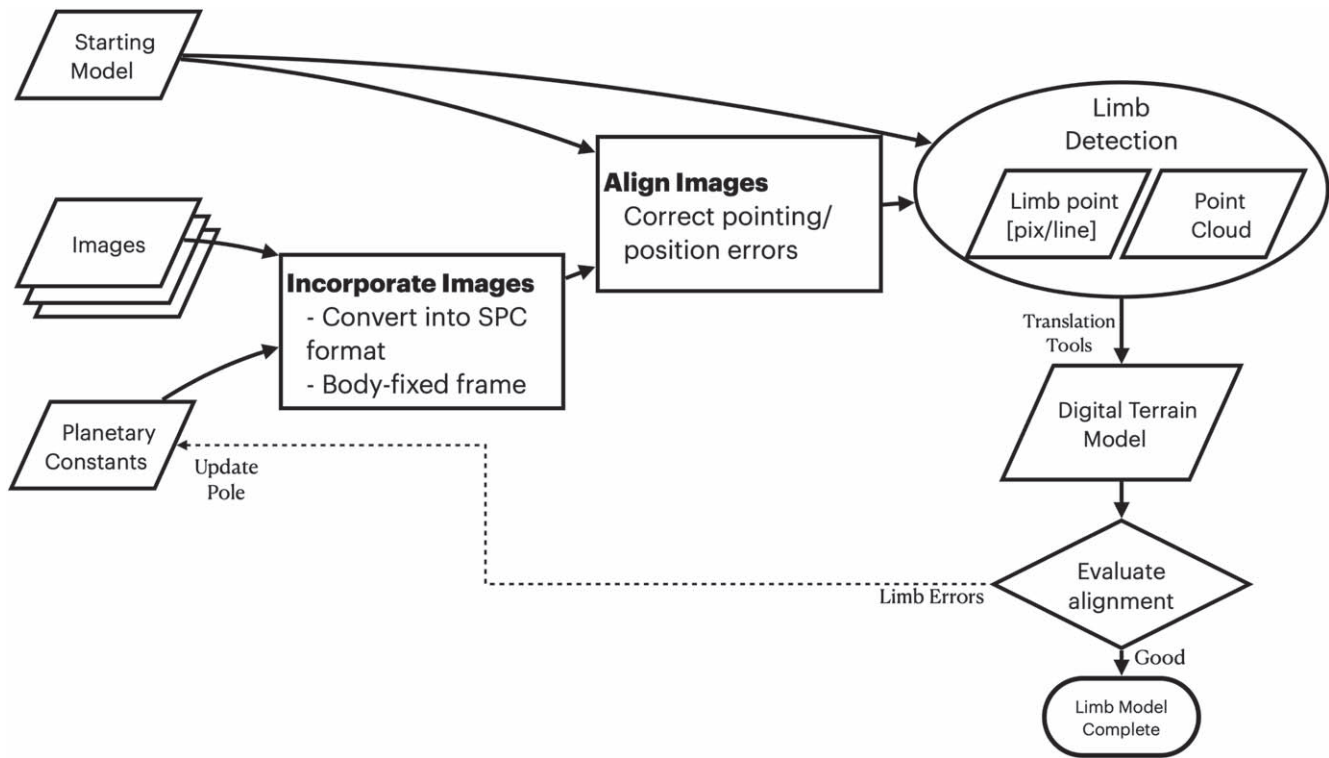


Figure 2. Generation of a DTM via limb-based modeling using data obtained during approach to the target. Once the images are incorporated (converted into an SPC-compatible format) and aligned to the existing DTM, the limbs are extracted. The limbs are combined as a point cloud to construct an updated model. Translation tools convert or combine the point cloud of vertices to a DTM [VECS2SHAPE/VECS2CUBE].

The first step in the limb-based modeling process is to ingest the images into the SPC system. The details are described in Section 3.4.1. Once that is done, the limb points are extracted. The SPC program [LIMBER] identifies the image’s pixel/line (row and column) location for every limb point. It does this search by identifying when the DN, as traced from the center of figure, drops below a user-defined threshold within a search radius. It uses a threshold and search radius to mitigate errors due to image artifacts and cosmic-ray effects. Using the starting DTM as the basis of the coordinate system, LIMBER outputs vectors from the center of mass to the limb points. Another tunable parameter is the distance between the identified limb points. Using larger distances results in a lower-resolution model, but this model can be a more appropriate representation of the overall shape. Too small a distance between points can create artifacts and greatly increase processing time. As the model and data improve, increased resolution can be assimilated into the models. If imaging at higher resolutions is limited to certain regions, the previous lower-resolution model point cloud can be used to fill in the gaps in the improved model.

The limb points are translated into Cartesian coordinates that are set in a plane that goes through the center of mass orthogonal to the line of sight. As the body rotates, each image allows the determination of a new plane of points that sweeps the entire surface (assuming a complete set of images covering a full body rotation). When all of the images are processed, the outcome is a set of vectors that define a 3D point cloud of limb points in a body-fixed Cartesian space.

This resultant point cloud becomes input to a program that generates a DTM [VECS2SHAPE/VECS2CUBE], for which SPC uses the implicitly connected quadrilateral (ICQ) format

(e.g., Gaskell et al. 2008) to define the plate model. The ICQ format is a list of vertices in a body-fixed frame that avoids needing a list of facets or plates by requiring the vertices to follow a specific order. This model is then fed back as a updated starting point for additional refinement of the limb-based model. The additional refinement includes adjusting the spacecraft’s camera position and pointing to improve the alignment of the images to the DTM. The whole process is repeated until the limbs of the rendered DTM converge with those in the images. We look for the differences between the two sets of images to become small, such as less than a few pixels.

Additional images can provide additional unique vectors to the solution because they are likely to be out of phase with previously obtained image sets and thus sample unique limb points. This phase offset improves the rotational coverage of the image set and thereby the additional vectors that define the limb-based point cloud.

In limb-based models, large convex features (such as boulders or ridges) are easily modeled. Concave surfaces, however, cannot be modeled because they will never be viewed on the limb. If the concave surface is large, it creates a source of error in the DTM. Maplet-based SPC modeling, however, does not have this limitation and is able to create a DTM at any GSD with sufficiently high image resolution.

For limb-based modeling, not only must the limb be in the image, there must also be a sufficient amount of the object to register the images to the DTM. As the spacecraft approaches the asteroid, the resolution improves until a point where the images can no longer be sufficiently registered to the DTM. At that point, one can transition to what we call the core SPC process: maplet-based modeling.

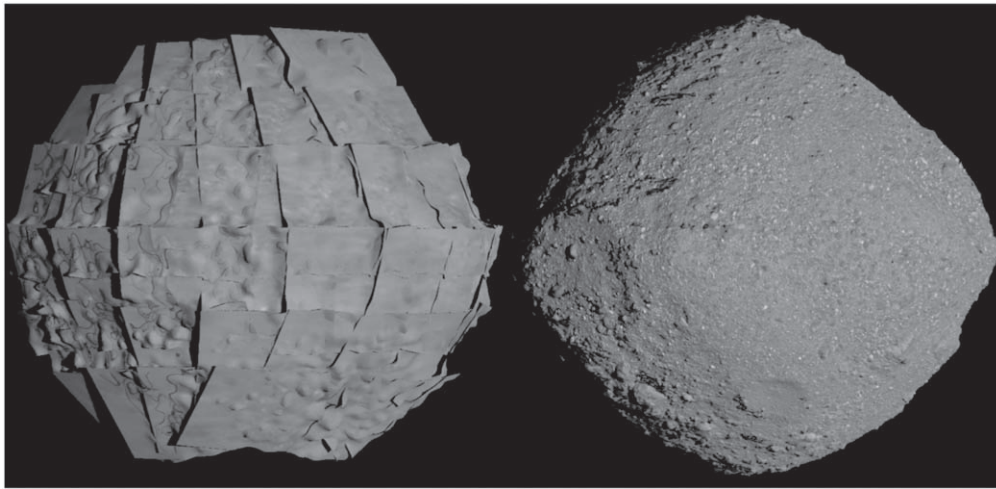


Figure 3. During maplet-based modeling, the surface is covered with small maplets. The image on the left shows the starting state of a global set of maplets with only the most basic generation of topography (see Section 3.4.2). As the data are processed and the DN information is converted into a consistent topographic solution, the maplets' topography changes to reflect the surface of the target. Eventually, the maplets will curve and make a closed and consistent object from which the data are resampled to make a global DTM.

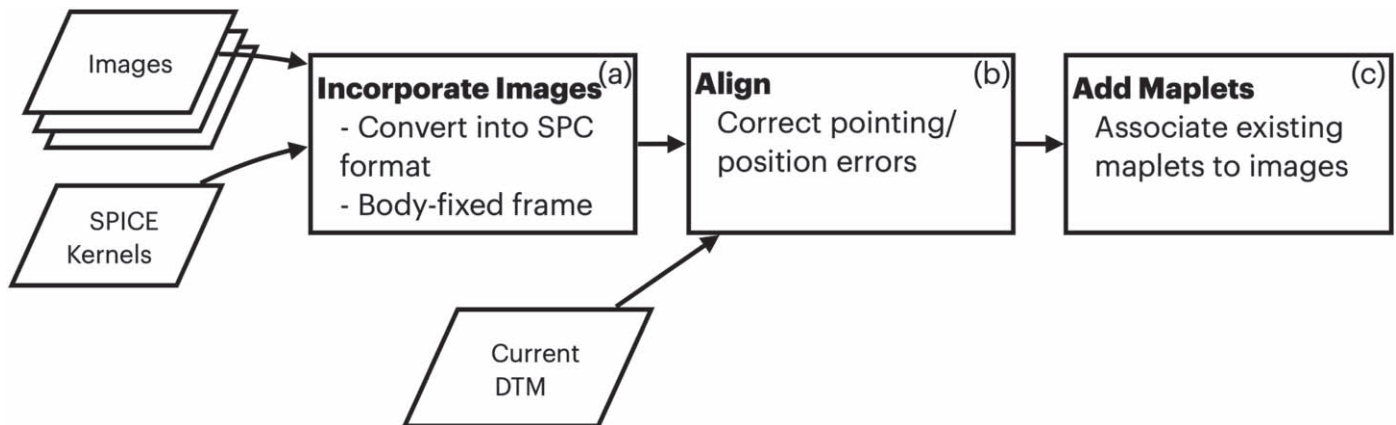


Figure 4. Ingestion of new images into SPC. (a) Images are converted into a simple binary 2D array of unsigned short integers, and the geometric reference system of the images is converted into a body-fixed frame. (b) The images are aligned with the existing DTM. (c) Previously existing maplets are associated with the images.

3.4. Maplet-based Modeling: Full SPC Process

The limb-based modeling discussed above provides a strong foundation for the generation of a DTM using the full SPC process. Maplet-based modeling provides a significantly improved ability for higher-resolution data, as well as supporting irregularly shaped objects. The process consists of creating maplets at higher and higher detail (Mastrodemos et al. 2011, 2012, 2015). The process, known as tiling, involves generating increasingly high-resolution maplets from increasingly high-resolution images as the spacecraft gets closer to the target and new detail is resolved (Figure 3). Tiling and the subsequent processing is the core SPC process that is used to gradually improve the DTM. It repeats the following three steps (Figure 4).

Ingestion: incorporating new images;
Alignment, extraction, and solve (AES): extracting image shading to generate topography; and
Tiling: improving maplet GSD by assimilating updated topography into higher-resolution maplets (or models).

3.4.1. Ingestion

When new images are downlinked from the spacecraft, they are ingested into the SPC system using a mission-specific translator [PROCESS_FITS] that reads the source images (typically either Flexible Image Transport System, FITS (Pence et al. 2010), files or a 2D binary array) and extracts the image data and metadata needed by SPC (Figure 4(a)). The mission-specific ingestion program stores the image data as a simple binary 2D array as unsigned integers (16 bit big-endian). Additionally, the metadata are translated [MAKE_SUMFILES] into a separate file for each image (called NOMINALS and SUMFILES) that has the image time; position and pointing⁶ of the camera in body-fixed coordinates; a valid DN range; previous and following images; expected uncertainty; and

⁶ The position and pointing parameters are extracted from a suite of files known as SPICE kernels. SPICE provides both a data format for spacecraft navigation information and tools and libraries to process the kernels. These kernels provide all information needed to calculate the position and pointing of every element on the spacecraft and solar system objects (Acton 1996; Acton et al. 2018).

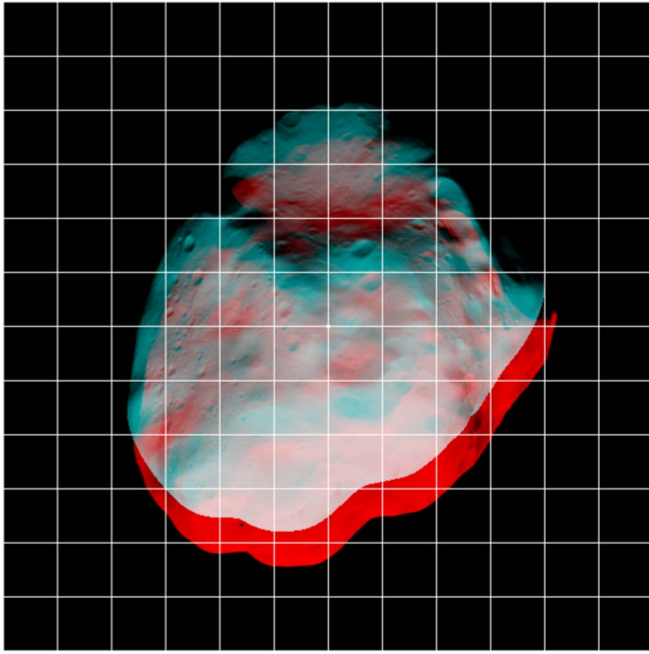


Figure 5. Actual navigation alignment offset for asteroid Lutetia, image N332048572, using reconstructed SPICE kernels. The real image is colored red. The rendered DTM image is colored cyan. When the actual image and the DTM agree, the alignment image is gray to gray-pink. We can easily identify a camera position/pointing shift that is required. Also, it allows interpretation of variations in overall size and large surface features. If there are deviations between the models, they can be seen as distinct shifts of features between the two channels.

residual errors. Typically, the images are raw images so that SPC can control all image calibration procedures and algorithms, but on some missions (e.g., the Lunar Reconnaissance Orbiter; Keller et al. 2016), calibrated images are used. Traditionally, SPC prefers to use raw images and do its own calibration so that the adjustments to the images are well understood and under the control of the SPC team.

The position and pointing of the spacecraft at the time of image collection are converted from the EME-2000 inertial frame into a body-fixed frame (Figure 4(a)). This means that the position of the camera is converted into a system where the center of mass of the object is the coordinate center, the x -axis is defined by the prime meridian, the z -axis is along the pole (right-hand rule), and the y -axis is orthogonal to the x - and z -axes (Gaskell et al. 2008). The position and pointing information comes from either the spacecraft telemetry, the predicted trajectory, or the navigation team’s reconstructed solutions and is used in a SPICE framework.

The next step is to align the images to the existing DTM [REGISTER] (Figure 4(b)). Even the best reconstructed spacecraft trajectory will have small offsets in the calculated position of the target when compared to its position in the images. Used to align the images to the DTM, SPC renders the DTM according to each image’s position, pointing, and solar illumination geometry. This rendered image is displayed side by side with the actual image to allow a visual check of alignment. Additionally, the two images can be superposed in different color channels (Figure 5).

Once the registration is completed, the new images are added to the existing maplets (Figure 4(c)). This batch adding is done via a batch process [AUTOREGISTER] where every maplet that falls within the image footprint is added. A culling step is

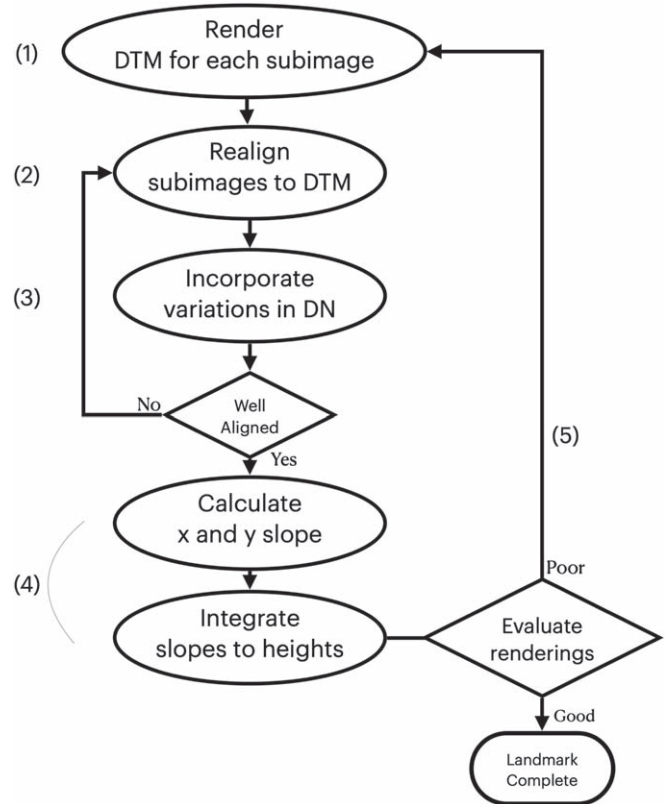


Figure 6. AES is the multistep process of extracting the DN values from the images to correct the DTM by comparing them with the rendered model’s DN values. Each step improves the topography and albedo, which results in a more accurate representation of each model subimage. An example subimage pair is shown in Figure 7. These steps are all accomplished in the module called LITHOS.

conducted where maplets that have poor viewing geometries or poor agreement with the image are rejected. Typically, we exclude maplets where the images have a more than 60° emission angle, maplets that have only an edge of the image, or the image’s pixel size was more than a factor of 4 lower than the GSD. However, in a data-poor environment, these parameters can be expanded to increase the amount of data.

3.4.2. AES: Converting Shading to Height

The term AES is used for the most in-depth step in SPC, where surface shape information is extracted from images to generate a DTM (see R. W. Gaskell et al. 2022, in preparation, for a detailed mathematic description). This process is done extensively throughout the DTM building process [LITHOS]. It is repeated whenever new images are added or higher-resolution maplets are produced and until there is a well-converged solution (Figure 6).

Unlike stereo and lidar techniques, the SPC process is iterative. It generates small corrections that are added to the current solution. The process of adding small corrections is continued until the corrections become small enough that additional processing provides no significant change in solution. The center position of the maplet is calculated through stereo; all other points within the maplet are derived from multi-image 2D photogrammetry with a variety of conditioning parameters (Gaskell et al. 2008). Each maplet is processed in turn as follows.

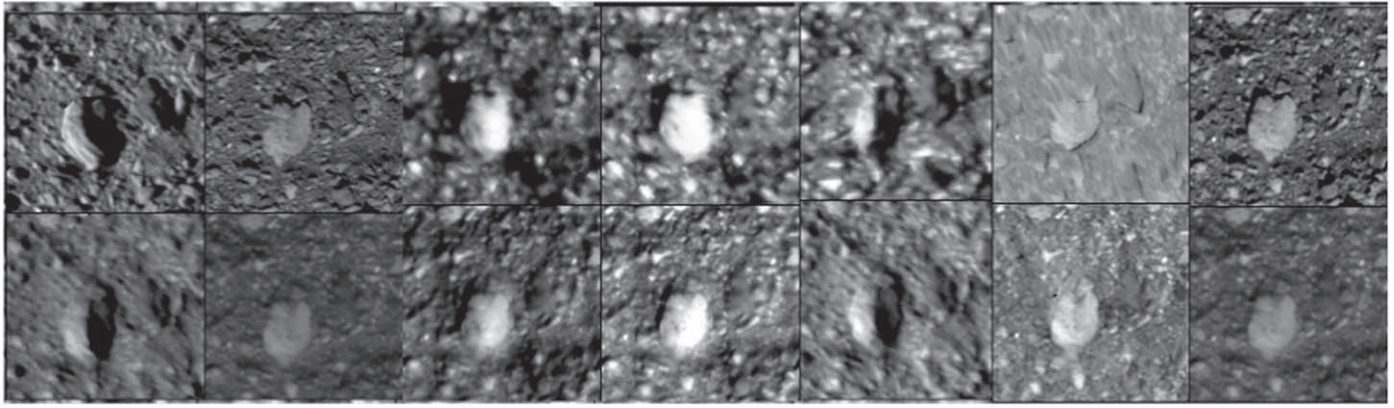


Figure 7. Example of a key processing output from SPC, the landmark display image. The top row shows source subimages, which are 99×99 pixel croppings of the original images centered at the landmark’s position. The bottom row shows the model subimages, that is, the DTM and albedo rendered with the same observing geometry as the associated real image. The agreement between each pair of source and model subimages indicates how well the DTM represents the actual topography and albedo.

Step 1: render DTM for each subimage. Each source image is resampled by SPC to form a “source subimage” (Figure 7, top row). The source subimage is orthorectified at the maplet’s GSD and pixel dimensions. These source subimages provide the source data for the maplet’s topography. For each of these real subimages, SPC renders a corresponding “DTM’s subimage” using the current DTM with the same observing geometries (Sun and spacecraft position) as the real subimage (Figure 7, bottom row). Each grid element of the subimage, or predicted brightness model (Mastrodemos et al. 2015), is based upon every image’s reflectance calculated from its scaled albedo value, the relative slope to the Sun and the photometric function. Several photometric functions can be used by SPC, but the default is a lunar Lambert function (McEwen 1991; Equation (1)). The term β is defined as the emission angle divided by 60° , u is the cosine of the emission angle, u_o is the cosine of the incidence angle, and L is the brightness function. Differences between the source and model subimages provide a measure of how closely the model fits the image and, by assumption, the actual topography, details described in Gaskell et al. (2008):

$$f = (1 - \beta)u_o + \frac{2Lu_o}{u_o + u}. \quad (1)$$

If the DTM lacks sufficient detail, its subimages will not match the corresponding source subimages (Figure 7). This lack of quality can be evident as fuzzy terrain, missing surface features, or features not aligned. These types of deviations are indicators that additional processing is needed on the model, so the extraction process is continued until the deviation becomes small.

Step 2: realign subimages to DTM. As the maplet’s DTM is improved, the subimages can be more accurately aligned with respect to one another. An automated SPC routine is used for a cross-correlation analysis to identify small shifts that improve the match between the source and model subimages. If the automated routines do not work, a manual shift can be applied. This change refines the pixel and line location of the feature within the image.

Because SPC can incorporate images from any illumination and observational geometry, it allows a greatly expanded amount of data available for DTM generation compared to other techniques. When the source subimages are generated, they are all orthographically projected (as if viewed from directly above), which results in removing the variations caused by observational geometry. As a side function to the AES process, SPC can, and usually does, update the spacecraft camera’s position and pointing; see Section 4.2.

Additionally, each image will have different illumination geometry; therefore, each source subimage will have different shadows and shading. As such, source subimages are difficult to align with one another. Shadows and shading variations can create offsets resulting in incorrect alignments. Typically, SPC uses a more stable process in which each source subimage is aligned with its corresponding DTM subimage. Each subimage pair will have the same illumination geometry: one actual image and one rendered with a matching Sun position. Aligning each source subimage with its corresponding DTM subimage ensures correct alignment among all images, even for disparate lighting and viewing conditions.

Step 3: incorporate variations in DN. As an improvement to the orthorectification of the images, SPC uses the difference between the pair of source and model subimages. A small amount of the difference between each pair of subimages is added or subtracted to a temporary template. The amount is user-controlled. This temporary template provides the DN that is used to calculate the x - and y -slopes for every pixel.

Step 4: calculate x - and y -slopes and convert to height. The key to SPC’s DTM generation is its calculation of heights from a 2D array of x - and y -slopes. It integrates the 2D slope field into a 2D array of heights relative to the normal plane, the mathematics of which are described in R. W. Gaskell et al. (2022, in preparation). We can use additional constraints to ensure a stable solution. The possible constraints for SPC are as follows.

(a) Current global DTM. When there are missing data in the images, the current global DTM can be used to fill

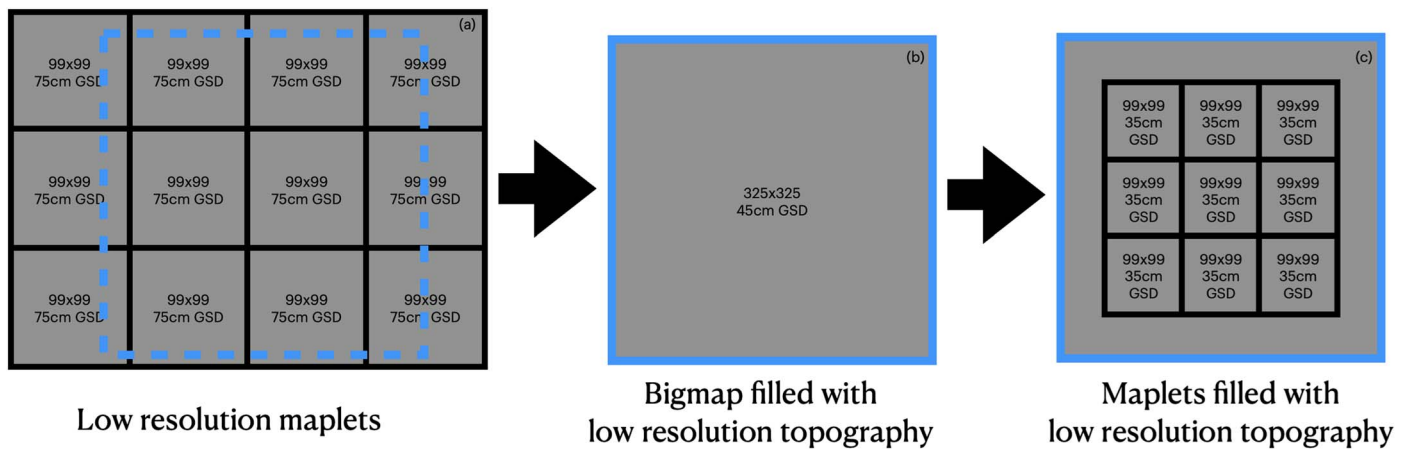


Figure 8. The process of increasing the resolution via tiling is as follows. (a) The current set of maplets covers a region at a specific GSD, in this case, 75 cm. (b) We create a bigmap (blue box) that resamples the current set of maplets at a GSD 30% larger than the desired GSD; desired is 35 cm, so the bigmap has a GSD of 45 cm. (c) A new set of maplets with the desired GSD is created and loaded with topography from the bigmap as a starting point, then new higher-resolution images are associated with these new maplets. The AES process is started with these new maplets in order to improve their topography. For visual clarity, this figure depicts a general example, specifically that panels (a) and (c) show the maplets touching one another. In practice, there is a 30% overlap between maplets.

the gaps. This option is useful when the maplet GSD gets so small that there are gaps in the image coverage.

(b) Overlapping maplet heights. During processing, neighboring maplets will partially overlap. This option, which is typically used, allows the heights from overlapping maplets to be added as an additional constraint during the conversion from slope to height.

(c) Heights of maplets as detected in limb images. If any portion of a maplet is located on the limb of an image, its radius from the center of mass will be used as a constraint.

(d) Reference DTM. If used, SPC will take the existing global DTM as a reference. These reference height positions will become a weighted input to the height calculation. This constraint allows SPC to reduce the deviation away from that reference model.

(e) Stereo calculations. Taking a random 1% of the points in the maplet, SPC will use stereophotogrammetry to calculate the height at that point. The heights at these points are used in the integration of the slope to height calculations.

(f) External DTM. One can specify an arbitrary DTM (such as topography from laser altimetry or stereo) as a constraint. Taking a random 1% of the heights of this external DTM, SPC will use them in the integration of the slope to height calculations.

Step 5: evaluate renderings. When the above steps are complete, the DTM is saved to the maplet file. The subimage pairs are rerendered, and the process is repeated again until the user determines that the solution has reached the point of diminishing returns. Testing has shown that 10 iterations of the AES process provides a stable solution (E. E. Palmer et al. 2022, in preparation).

3.4.3. Tiling: Improving Maplet GSD

In SPC, tiling is the process of covering the surface with maplets at an increased resolution [LITHOS] (Figure 8). The higher-resolution (smaller-GSD) maplets allow for a finer sampling of the image data.

The first step for increasing the resolution is to resample the existing maplets (Figure 8(a)) to form a reference DTM [BIGMAP]. Typically, this reference DTM will be a bigmap (Figure 8(b)), but it could be a global DTM when working with low-resolution data. The existing maplets have a coarser GSD than the ultimately desired GSD but cover the entire region of interest.

Using this bigmap, we create new, finer GSD maplets that cover most of the bigmap. The bigmap should be larger than the size of the final region. The new maplets can be defined using either a latitude/longitude pair, an x/y position in the bigmap, or a pixel/line location within an image.

In the example shown, Figure 8, the maplets have a GSD of 75 cm, and we want to generate terrain at 35 cm GSD. As a technique, we create a bigmap that has a GSD that is 30% larger than the desired maplets. This allows the generation of maplets that are 99 pixels wide and positioned every 50 bigmap pixels, which results in a 30% maplet overlap.

Each new maplet has a defined position in 3D space, a GSD, and a pixel width, but it needs starting topography. We use the topography of the bigmap as the initial topography of the maplets. Although the resolution of the topography is less than that of the maplet GSD and new images, it is sufficient to start the process.

Once the topographic fill of a maplet is done, SPC adds the images to each maplet that fall within that maplet's footprint [LITHOS]. A basic step of culling is done to remove each image that has too high of emission or illumination angles, too little of the maplet within its footprint, or resolution that is too poor. In practice, maplets with images whose emission angle is greater than 60° are removed.

When there is a suite of good imagery (Section 4.1), the automated tools work without much manual effort. Because there is existing topography loaded into the maplet, albeit lower in resolution, there are little issues with aligning the real subimages to the model subimages. Usually, this process can be done via automated tools in batch; however, sometimes human effort is needed to better align the source subimages, add more images, or remove unusable images.

Once these steps are complete, the AES process (see Section 3.4.2) can be conducted. Because the GSD of the

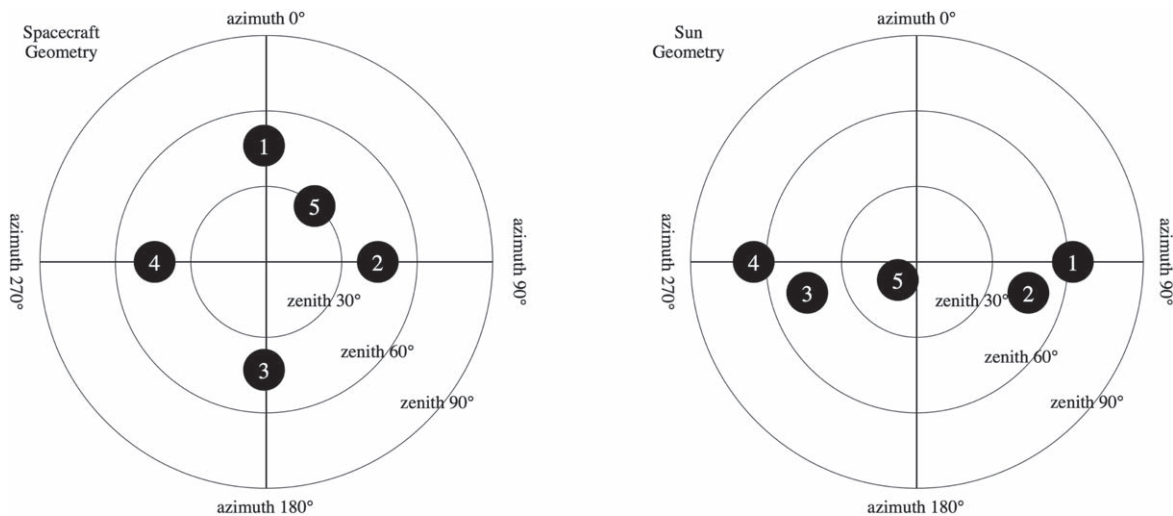


Figure 9. Example of a set of high-quality images for a piece of terrain. The left panel shows the position of the spacecraft as seen from the surface, looking up. The right panel shows the position of the Sun from the surface. Azimuth runs from north (0) to east (90) to south (180) to west (270). Zenith angle is marked every 30°, with zero being straight up. This figure depicts an asteroid with a near-zero obliquity and the point of view of a maplet that is north of the equator by 20°–40°. Each image station is denoted by a matching pair of numbered circles in both the spacecraft and sun geometry figures. Topographic images are numbers 1–4, with the albedo image being number 5.

new maplets is decreasing, there are more pixels per physical area, and more of the image data can be stored. Generally, the image’s pixel size is smaller (higher-quality) than the GSD of the new maplets. Typically, SPC can easily extrapolate the data by a factor of 2. When needed, such as when data are scarce, SPC might be able to extrapolate the images by a factor of 4, meaning that the pixel size can be four times the GSD of the maplets. The AES process will pull whatever additional image information exists into the new maplets.

The AES process is typically an iterative process, where the increase in resolution results in a more accurate topography that allows the images to be more accurately aligned. Iteration of the AES process continues until the solution converges relative to a predefined criterion.

3.5. Finishing the Model

Global or regional models can be generated at any point throughout the SPC process by resampling all of the maplet data [DENSIFY/BIGMAP]. The model will reflect the quality at that specific point in development and can be used for error detection.

In practice, temporary models are created and used to evaluate the entire system for internal consistency and are a key part of SPC and its techniques. Incongruous image alignments, which can be corrected manually, are detected and reported by SPC. If the errors in the spacecraft’s reconstructed positions are large, then updates to the spacecraft’s position and pointing are made to the entire suite of images to minimize the residual error [GEOMETRY/DYNAMICS]. Errors in the DTM’s heights can be detected [RESIDUALS] by residual errors between maplets that cover the same region, again allowing for manual correction. Additionally, as a final check, SPC can evaluate the maplets to ensure that as many of the images are used in as many maplets as possible. Final checks are made to minimize errors from uncertainties in the navigation solutions obtained, poor handling of shadows, or low number of images in maplets.

After all images have been incorporated, the desired GSD maplets have been generated, and any errors, lack of data, or inconsistencies are fixed, a DTM will be generated. The DTM

generation is done by resampling the maplets to form a larger (regional or global) DTM. Each vertex receives a height value that is the weighted average of all of the maplets at that latitude and longitude (i.e., the maplets with the smallest GSD/highest resolution affect the solution of the final product more than the maplets with larger GSD). These final values are stored either in a global DTM using the ICQ format or as a bigmap. These files can be converted into any other DTM format as needed.

4. Discussion

4.1. Mission Planning

Using SPC for developing a topographical model benefits from implementing an SPC-compatible imaging plan during mission operations. While we present an optimum suite of image observation conditions, SPC has proven to be quite resilient with suboptimal data sets; however, the quality of a DTM will be higher when the input images were planned with SPC in mind. Most imaging plans start with a distant approach (Section 3.3.1). As previously discussed, this phase is important to update the starting model for the SPC process and better estimate planetary constants (rotation rate and pole orientation). A carefully planned imaging campaign during approach can, with the addition of Doppler tracking information, greatly increase the knowledge of the size of the target. These approach-phase constraints, which provide the basis for the limb-based starting model, are modest when compared to those required for high-resolution DTMs. Although a DTM can be generated using SPC from as few as three images, the quality of the DTM is strongly affected by the image viewing geometries (Barnouin et al. 2020).

E. E. Palmer et al. (2022, in preparation) used a simulated data set to assess how the number of images and the quality of the observation influence the quality of an SPC-derived DTM. This analysis informed the OSIRIS-REx imaging strategy. The results indicated that a strong imagery set that supports DTM generation is a total of five images that have different observing geometries for the terrain to be modeled. Four of the images focus on topographic information, while the fifth focuses on

albedo. While the general term of “emission angle” is heavily used for spacecraft imaging requirements, it is insufficient to specify the full requirements in 3D space desired for SPC. Instead, it is important to break down the observation into a two-component vector (Figure 9). One component is the azimuth of the vector from the target to the spacecraft. The other component is the solar zenith angle, the angle of the vector to the spacecraft measured from the target’s zenith position.

Testing has shown that a series of four images, with the azimuth angle roughly orthogonally north, south, east, and west of the region, is best for maximum topographic determination and shadow mitigation. The exact orientation of the azimuths need not be cardinal directions, but the ideal variation in azimuth angles would be near 90° , with 60° as a practical minimum between any two observations. The range of acceptable zenith angles can be broad. For operational use, zenith angles between 20° and 45° are effective. The topographic images (1–4 in Figure 9) provide optimum viewing geometries to extract the slope in the north–south and east–west directions. Incidence angle is not a critical component for topographic images, but extreme angles ($>75^\circ$) greatly reduce the images’ usability.

Approach imaging campaigns frequently have a series of images taken every 10° – 20° of rotation as the spacecraft approaches the target. Such observations limit the accuracy of the DTM for solving for north- and south-facing slopes. Nevertheless, the quality of the DTM with the limited approach geometries would be sufficiently good for low- to moderate-resolution science assessments of the target shape and provide a useful foundation for further DTM generation. When possible, getting images from the north/south directions will reduce topographic errors in north- and south-facing slopes.

The albedo image needs to be acquired when the Sun is as near local noon (low incidence angle) as possible without getting near zero phase angle (typically $<7^\circ$) to avoid low-phase opposition effects. The azimuth angle is not a critical component for albedo images. When the Sun is at local noon, the topographic signature is minimized, and the albedo signature is maximized; these are the conditions captured in the fifth image (Figure 9). By having data that uniquely emphasize the albedo of the surface, SPC is able to determine how much variation in an image is controlled by topography versus albedo.

Getting a complete set of observation geometries can be difficult. Besides complex mission planning, many objects have near-zero obliquity, thereby limiting the range of solar azimuth angles. For such objects, equatorial regions will only possess east and west azimuths with no north or south components. At higher latitudes, the range of azimuths can include either a north or south solar azimuth component, but these same high-latitude areas never gain the opposite solar azimuth and suffer from long shadows that prevent the highest-quality modeling. Other difficulties include 90° obliquity or regions that are in permanent shadow.

Objects with obliquities larger than a few degrees, such as 67P C-G, can provide opportunities for wider illumination geometries at the cost of a much longer arc of observations because this variation is expressed over the course of half an orbit. Likewise, tumbling asteroids provide these opportunities at the cost of additional complexity in mission planning and rotation state estimation.

Although the above text specifies an optimum suite of images for DTM generation, the reality is that SPC is able to effectively use a broad range of images in the generation of its maplets. Not only is the range of illumination and observation geometries broad, but even the resolution of the images can vary widely. Because each image is registered to better than 0.1 pixel, and each image’s pixel boundaries fall in slightly different positions, bicubic interpolation can be used to extract information below the pixel level in images that have a pixel size larger than the GSD of the maplet. For SPC, the practical limit for lower-resolution images is a factor of 4 (i.e., if making a 1 cm GSD maplet, it is best to exclude images with a pixel size >4 cm.) When both are included in a maplet, SPC weights higher-resolution images more than lower-resolution images.

A typical rendezvous mission progression consists of improving the resolution of the imagery and thereby improving the GSD of the model. For planning purposes, experience indicates that as the next higher-resolution images are added into SPC, their resolution should increase by a factor of about 3. For example, images with a pixel size of 5 cm would generate a good-quality DTM with a GSD of 5 cm, and a comfortable step in resolution for the next set of images would be a pixel size of 1.6–2 cm. Images where the resolution jumps are wide are harder to register to the existing maplets and can result in the automated tools failing, requiring extensive manual processing. For planning, imagery jumps of a factor of 10 in resolution should be avoided because they increase the amount of manual work that is required to process them.

4.2. Supporting Spacecraft Navigation

Once SPC has generated a stable model, the topography and, more specifically, the landmark positions can be used to calculate corrections to the spacecraft position and pointing [GEOMETRY]. This update process is similar to the bundle adjustment (Edmundson et al. 2012) that is done in stereophotogrammetry, where the stereo solution provides updates to the position of its source data. For SPC, this process is significantly more integrated and expansive, impacting all of the images as a cohesive system.

The initial position for the camera in inertial space comes from the navigation team’s predicted or, if available, reconstructed flight trajectory. These data are typically delivered as SPICE kernels. However, the position and pointing of the spacecraft camera can be improved in lockstep with DTM improvements.

For missions that orbit the target, most images are assigned to more than 50 maplets, which results in a large number of control points (landmarks) that can be used for this update. For every image, SPC will register the pixel and line location of the center of each maplet to around one-tenth of an image pixel. Projecting the control points on the image back into space, it is possible to calculate the position of the spacecraft when the image was taken. Additionally, because there are more than the minimum three control points in the system (frequently over 50), there is an overconstrained solution. Because all control points should converge to the same location, the dispersion of all of the data is an indication of how well topographic, spacecraft position, and spacecraft pointing errors have been removed.

The ability of SPC to update the position (and pointing) of the spacecraft and the topography creates a potential degeneracy unless other constraints are included. Because the 3D

Table 2
DTM Products Generated Using SPC for Progressive Observational Phases (Lauretta et al. 2021) of the OSIRIS-REx Mission

Mission Phase	Spacecraft Position	Number of Images	Image Pixel Size (cm)	Product	Product GSD (cm)	Version
Distant Approach	Equator	5 sets of 4 images ^a	>10,000	Center-finding DTM ^b	N/A	
Approach	Equator	6 sets of 18 images ^c	2000–100	Limb-based DTM	300	
Close Approach	Equator	424 images ^{c,d}	100–50	Approach DTM (draft)	150	v10
Preliminary Survey	Poles and equator ^c	683 images	70	Navigation/science DTM	75	v20
Baseball Diamond	S to N ^e	9391 images	5	Final science DTM	35	v42
Recon A	Overflight ^f	324 images	2	Navigation features ^g	3	
Recon B	Overflight ^f	366 images	1	Navigation features ^g	0.5	
Recon C	Overflight ^f	345 images	0.25	Navigation features ^g	0.5	

Notes.

^a Imaging a single side of Bennu.

^b Validate starting model's size via center finding.

^c Suite of images taken every 20° of rotation.

^d Due to uncertainty in the asteroid's position, 2×2, 3×3, and 4×4 mosaics were acquired.

^e Camera pointing was nadir and off-nadir slewing.

^f The spacecraft flew over candidate sample collection sites at reduced altitudes and performed multiple scans.

^g Rendered features supporting autonomous navigation were added to the mission plan when suitable sample collection sites were smaller than expected, forcing a reliance on autonomous navigation for the sampling event (Olds et al. 2022; Norman et al. 2022).

positions of the maplets are calculated from the images, and the position and pointing of the images are derived from the maplets, the system could drift, in which case the target's size and distance change systematically. To avoid this, SPC uses additional constraints to ensure that it does not result in an erroneous, albeit self-consistent, solution. The primary constraint used is the nominal (or expected) position and pointing provided by the navigation team and an SPC-derived uncertainty. When images are ingested, the uncertainty in their solution is stored. When SPC updates the position and pointing, it constrains the updated position solution based upon the nominal values (the starting position and pointing) and the uncertainty.

Additionally, images that contain the target's limb provide a strong constraint on SPC's distance and pointing solutions. If an image has the limb in it, any updates to the spacecraft position will be small compared to the distance to the target, so the angular size of the target remains effectively constant. With the angular size of the target nearly consistent and the distance changing little, the overall size of the target can be determined with a high degree of accuracy. As such, the size becomes a known parameter that ensures the distance remains consistent. Finally, any maplets that fall along the limb region provide orthogonal constraints on the pointing of the camera.

Finally, SPC constrains the solution by using the fact that the spacecraft's trajectory is smooth and consistent between maneuvers. Because the spacecraft's trajectory is smooth, SPC can assume that the camera position will follow the arc. Any deviation between the nominal and actual trajectory will be consistent among all images taken during that arc. When an image's camera position is updated, SPC algorithms weigh the deviation from the nominal four images before and after and limit the total deviation allowed.

As described in Section 3.1, maplets are added to images or images to maplets multiple times to ensure that as many registrations as possible are made. Each registration (match of the image pixel/line to maplet location) creates an equation with three unknowns (slope x , slope y , and albedo). When there are more than three registrations, which is typical, the system is overconstrained, allowing the calculation of a residual error, calculated by the module [RESIDUALS]. The residual error for

an image is the measured deviation between the pixel/line location of the maplet's center in the image (based upon cross-correlation image matching) and where the maplet's center is calculated to be (based upon geometric calculation). These data provide information to the navigation team about the uncertainty of the position of every maplet, as well as the updated solution for the image.

The feedback of the camera position and pointing to the SPC system allows for navigation errors to be reduced without output provided by module [REGRES]. As the navigation errors are reduced, it improves the projected position of the maplets in 3D space, resulting in a more accurate DTM. The extraction process continues until the changes in the positions become negligible (or stabilize). Once that is done, the DTM will have reached a high level of fidelity. A significant number of tools exist to review the DTM for specific and systemic errors and are covered in E. E. Palmer et al. (2022, in preparation). More details of the interactions between SPC and spacecraft navigation are provided in C. D. Adam et al. (2022, in preparation).

4.3. Representative Results: OSIRIS-REx as a Case Study

During the OSIRIS-REx mission (Lauretta et al. 2017, 2021), the Altimetry Working Group (Barnouin et al. 2020) used SPC to generate the DTMs that were used by the flight dynamics team for navigation around Bennu (C. D. Adam et al. 2022, in preparation) and provided the ultra-high-resolution DTMs used on board the spacecraft during the descent and sample collection (Norman et al. 2022; Olds et al. 2022). During approach, the team generated DTMs with a low resolution of 3 m GSD, which matched the quality of the images. By the end of the mission, the entire asteroid was modeled with a 5 cm GSD global terrain and select navigation-focused regions with terrain down to 0.5 cm GSD.

4.3.1. Approach

The first images used by SPC were taken on 2018 November 8, during the mission's Approach phase, when OSIRIS-REx was 960 km from Bennu (Table 2). Although the pixel size of the images was 219 cm, they were sufficient to generate a limb-

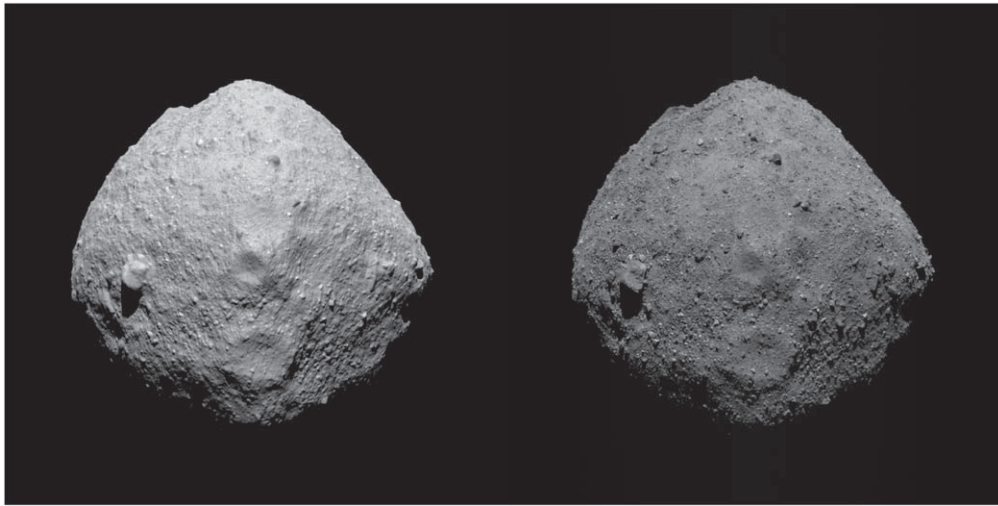


Figure 10. Comparison between a rendered image and an actual image of the OSIRIS-REx target, asteroid (101955) Bennu. Left is the SPC 35 cm (v42) DTM rendered with matching spacecraft position and pointing. Right is a MapCam image from the Preliminary Survey mission phase, image 20181213T022027S610. The orientation is based upon how the spacecraft imaged Bennu, and in this case, the north pole was facing to the right. The polar radius of Bennu (in this image, left to right) is 508 m.

based model at 300 cm GSD. Over the next 2 months, the spacecraft approached Bennu centered near the equator with a low phase angle. At each imaging event, the data were ingested into SPC and an updated model was created. During this time, refinements to the pole position and rotation rate were made.

When the spacecraft reached 810 km distance, the uncertainty in the position and pointing of the spacecraft relative to the camera's field of view (in this case, the OCAMS PolyCam imager; Rizk et al. 2018) was significant enough that a mosaic of images had to be taken to ensure that all of Bennu would be imaged (i.e., rather than a single image, the field was imaged in a 2×2 grid that bracketed the full range of where Bennu could be located when accounting for position and pointing error).

On the last 2 days of Approach, 2018 December 1 and 2, the spacecraft obtained the highest-resolution images that it would get for the following 2 months. These data were taken from slightly north of the equatorial line because the spacecraft had begun its trajectory toward the north pole in preparation for the Preliminary Survey mission phase that consisted of polar and equatorial flybys (Lauretta et al. 2021). The pixel size of these images was 32 cm pixel^{-1} . The ensemble of images allowed the Altimetry Working Group to generate a draft SPC model to be used by the science team for their initial analysis and the navigation team to start to check that the model was of high enough fidelity for navigation. This model, v10, consisted of 236 maplets and 2351 images with a GSD of 150 cm, and it had a formal uncertainty of 1.7 m. Formal uncertainty is discussed in Weirich et al. (2022) and R. W. Gaskell et al. (2022, in preparation).

4.3.2. Preliminary Survey

The Preliminary Survey phase continued the work from Approach. Images collected by the MapCam imager of OCAMS were incorporated into the DTM from two observation stations during a north pole pass, two stations from an equatorial pass, and two stations from a south pole pass (Lauretta et al. 2021; Table 2).

The incorporation of the images taken over the poles and the medium-to-high phase angle equatorial pass led to

improvements in the 75 cm maplets. Additionally, we created 35 cm maplets to enhance the accuracy of the topography. Testing had shown that generating topography at a smaller GSD than the final product GSD leads to a more accurate DTM (Weirich et al. 2022).

Bennu has many sizable boulders whose slopes are high enough to make SPC modeling difficult. During this effort with limited data, we modeled large boulders by creating DTMs with GSDs as small as 10 cm of focused area to aid in correctly determining the heights of the boulders.

The DTM produced at the end of Preliminary Survey had 1.6 million vectors, equating to a GSD of 75 cm. The data were sampled from a global mesh of overlapping 35 cm GSD maplets. This version of the model is called SPC v20 (Barnouin et al. 2019) and consisted of 2369 maplets and 5114 images with a GSD of 75 cm; it had a formal uncertainty of 0.54 m and an accuracy of 1.06 m as calculated via key point differences (Al Asad et al. 2021). This model was used by the navigation team and provided sufficient information to transition from center-finding navigation to landmark-based navigation (Adam et al. 2019; Leonard et al. 2020), which allowed the mission to conduct the Detailed Survey.

4.3.3. Detailed Survey: Baseball Diamond

The Baseball Diamond portion of the Detailed Survey phase had seven south-to-north passes of Bennu: two obtained medium-resolution images (about 20 cm), and the remaining five passes acquired images at pixel sizes of 4.7–6.0 cm, except in the very high latitudes (Lauretta et al. 2021; Table 2).

During operations, six of the seven passes met the requirements for the imaging geometry needed for successful SPC modeling, while one pass, Flyby 2, did not. A winter storm impacted operations, making it impossible to update the predicted spacecraft position and timing, which resulted in an offset of several hundred meters. This pass was to provide medium-resolution images that would allow intermediate GSD maplets to be made (14 cm) before the $\sim 5 \text{ cm pixel}^{-1}$ images were ingested. Fortunately, later passes provided sufficient data for our purposes, causing only a delay in processing. The highest-resolution images were sufficient to generate the final

SPC global model for scientific use at 35 cm GSD that was resampled from 14 cm maplets over Bennu except at the highest latitudes, as well as numerous higher-resolution DTMs required for navigation to the sample collection site (Norman et al. 2022; Olds et al. 2022).

This model (Figure 10), SPC v42, consisted of 14,814 maplets and 15,325 images with a GSD of 35 cm; it had a formal uncertainty of 0.19 m and an accuracy of 45 cm (Al Asad et al. 2021). This model was provided to the navigation team and provided sufficient accuracy for the mission to conduct the Reconnaissance (Recon) A, B, and C passes and support the sample collection rehearsals and event (Lauretta et al. 2021). More information about the performance of the DTM is provided in Leonard et al. (2020) and C. D. Adam et al. (2022, in preparation).

4.3.4. Regional High-resolution DTMs

Although global high-resolution images were available (5 cm pixel^{-1}), there were no mission requirements for a global model at finer GSDs than 35 cm. As such, once DTM v42 was created, the team focused on generating small regional DTMs that would be used for navigation during sample collection activities. Though not used operationally, SPC-based DTMs with 5 cm GSD were created. Data acquired by the OSIRIS-REx Laser Altimeter (Daly et al. 2017) were also used to create higher-resolution DTMs, which will be discussed in J. A. Seabrook et al. (2022, in preparation).

Once the global model was generated and the mission had selected candidate sites for collecting a sample, the focus shifted in the Reconnaissance phases (Lauretta et al. 2021) to building high-resolution local DTMs over those locations to provide the data required for navigation to the surface (Olds et al. 2022; Norman et al. 2022). We generated 176 local DTMs with sizes of 125×125 pixels at GSDs from 15 down to 0.5 cm. These DTMs were uploaded onto the OSIRIS-REx spacecraft and used for autonomous navigation using a technology called natural feature tracking (Olds et al. 2022). With the help of the localized DTMs, the OSIRIS-REx spacecraft navigated from orbit to the surface with two trajectory burns (Checkpoint and Matchpoint) and touched down within 74 cm of the desired location (Lauretta et al. 2021; Olds et al. 2022).

5. Conclusion

Through spacecraft imagery alone, the powerful SPC technique can provide DTMs of planetary targets at global and local scales of sufficient quality to be used for science (Barnouin et al. 2019) and navigation (C. D. Adam et al. 2022, in preparation; Olds et al. 2022). With high-quality data, SPC generates DTMs with an accuracy that is similar to the highest-resolution images available (E. E. Palmer et al. 2022, in preparation).

The iterative process of SPC requires some level of manual processing to supplement automated processing in difficult situations. Low-resolution models can be created in less than a day, but higher-resolution models (greater than 2 million plates) can require weeks to months to fully process (Weirich et al. 2022).

We suggest that coverage and image resolution be carefully considered during mission planning of spacecraft and solar geometries. One can use SPC for most any solid planetary

body, and it is able to process and use any images taken of that object regardless of viewing geometry. An optimal trade-off between DTM quality and mission resources is having SPC use five images (four optimized for topography, one optimized for albedo) with specific solar to target to spacecraft geometries. We provide this overview of SPC as a starting place for mission planners and future SPC operators to understand the capabilities, limitations, and requirements of SPC.

This material is based upon work supported by NASA under contract NNM10AA11C issued through the New Frontiers Program and by the Canadian Space Agency. We are grateful to the entire OSIRIS-REx Team for making the encounter with Bennu possible. All data from the OSIRIS-REx mission are available via the Planetary Data System at <https://sbn.psi.edu/pds/resource/orex/>. Digital terrain models of Bennu are available via the Planetary Data System and the Small Body Mapping Tool at <http://sbmt.jhuapl.edu/>. Licensing of the PSI SPC software suite is available from the Planetary Science Institute at <https://spc.psi.edu>. Please contact Eric Palmer (epalmer@psi.edu) for information on access, training, and the fee schedule.

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